ROCSCIENCE

Case4_Iter1_Simple
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:34:42 am
Software Version: 9.041

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Slide2 Analysis Information

Case4_Iter1_Simple

Project Summary

File Name:	Case4_Iter1_Simple.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:00.154s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:34:42 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	25
Tolerance:	0.0001
Maximum number of iterations:	100
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	5
Circles per division:	5
Number of iterations:	1
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	16 kN/m3
Cohesion	9 kPa
Phi	32 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	1.374660
Center:	24.151, 53.427
Radius:	28.855
Left Slip Surface Endpoint:	30.292, 25.234
Right Slip Surface Endpoint:	50.309, 41.247
Resisting Moment:	19240.8 kN-m
Driving Moment:	13996.8 kN-m
Total Slice Area:	51.7718 m2
Surface Horizontal Width:	20.0168 m
Surface Average Height:	2.58642 m

Method: spencer

FS	1.371240
Center:	24.151, 53.427
Radius:	28.855
Left Slip Surface Endpoint:	30.292, 25.234
Right Slip Surface Endpoint:	50.309, 41.247
Resisting Moment:	19192.9 kN-m
Driving Moment:	13996.8 kN-m
Resisting Horizontal Force:	523.4 kN
Driving Horizontal Force:	381.698 kN
Total Slice Area:	51.7718 m2
Surface Horizontal Width:	20.0168 m
Surface Average Height:	2.58642 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	38
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	33
Number of Invalid Surfaces:	5

Error Codes

Error Code -111 reported for 5 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-111 = Safety factor equation did not converge

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.37466

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.80067	2.90896	13.1048	Material 1	9	32	7.41402	10.1918	1.90721	0	1.90721	3.63316	3.63316
2	0.80067	8.57121	14.743	Material 1	9	32	10.1938	14.0131	8.02257	0	8.02257	10.705	10.705
3	0.80067	13.9186	16.3936	Material 1	9	32	12.7447	17.5196	13.6342	0	13.6342	17.3836	17.3836
4	0.80067	18.9433	18.0584	Material 1	9	32	15.0685	20.714	18.7463	0	18.7463	23.6594	23.6594
5	0.80067	23.6366	19.7391	Material 1	9	32	17.1663	23.5978	23.3614	0	23.3614	29.5211	29.5211
6	0.80067	27.9884	21.4378	Material 1	9	32	19.0387	26.1717	27.4805	0	27.4805	34.9561	34.9561
7	0.80067	31.9868	23.1564	Material 1	9	32	20.6853	28.4353	31.103	0	31.103	39.9501	39.9501
8	0.80067	35.6187	24.8974	Material 1	9	32	22.1052	30.3871	34.2266	0	34.2266	44.4862	44.4862
9	0.80067	38.8688	26.6634	Material 1	9	32	23.2965	32.0247	36.8472	0	36.8472	48.5454	48.5454
10	0.80067	41.7196	28.4571	Material 1	9	32	24.2565	33.3444	38.9593	0	38.9593	52.106	52.106
11	0.80067	44.151	30.2819	Material 1	9	32	24.9819	34.3416	40.5549	0	40.5549	55.1426	55.1426
12	0.80067	46.1397	32.1413	Material 1	9	32	25.4681	35.01	41.6247	0	41.6247	57.6264	57.6264
13	0.80067	47.6587	34.0395	Material 1	9	32	25.7098	35.3422	42.1563	0	42.1563	59.5236	59.5236
14	0.80067	48.6764	35.9812	Material 1	9	32	25.7002	35.329	42.1352	0	42.1352	60.7947	60.7947
15	0.80067	49.1557	37.9721	Material 1	9	32	25.4315	34.9596	41.5441	0	41.5441	61.3933	61.3933
16	0.80067	49.0524	40.0186	Material 1	9	32	24.8941	34.2209	40.3618	0	40.3618	61.2642	61.2642
17	0.80067	48.3132	42.1285	Material 1	9	32	24.0769	33.0976	38.5641	0	38.5641	60.3411	60.3411
18	0.80067	46.8735	44.3114	Material 1	9	32	22.9667	31.5714	36.1218	0	36.1218	58.5429	58.5429
19	0.80067	44.6532	46.579	Material 1	9	32	21.5478	29.6209	33.0004	0	33.0004	55.7698	55.7698
20	0.80067	41.551	48.946	Material 1	9	32	19.8018	27.2208	29.1593	0	29.1593	51.8954	51.8954
21	0.80067	37.4365	51.4316	Material 1	9	32	17.7069	24.3409	24.5505	0	24.5505	46.7566	46.7566
22	0.80067	32.1357	54.0612	Material 1	9	32	15.2369	20.9456	19.117	0	19.117	40.136	40.136
23	0.80067	25.4083	56.8704	Material 1	9	32	12.3619	16.9934	12.7921	0	12.7921	31.7339	31.7339
24	0.80067	16.9046	59.9108	Material 1	9	32	9.04694	12.4365	5.49951	0	5.49951	21.1131	21.1131
25	0.80067	6.07816	63.2638	Material 1	9	32	5.25544	7.22445	-2.84148	0	-2.84148	7.59139	7.59139

Global Minimum Query (spencer) - Safety Factor: 1.37124

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.80067	2.90896	13.1048	Material 1	9	32	9.48923	13.012	6.42056	0	6.42056	8.6296	8.6296
2	0.80067	8.57121	14.743	Material 1	9	32	12.5265	17.1768	13.0856	0	13.0856	16.3819	16.3819
3	0.80067	13.9186	16.3936	Material 1	9	32	15.1888	20.8275	18.9279	0	18.9279	23.3963	23.3963
4	0.80067	18.9433	18.0584	Material 1	9	32	17.5004	23.9972	24.0005	0	24.0005	29.7064	29.7064
5	0.80067	23.6366	19.7391	Material 1	9	32	19.4821	26.7147	28.3495	0	28.3495	35.3401	35.3401
6	0.80067	27.9884	21.4378	Material 1	9	32	21.1524	29.005	32.0147	0	32.0147	40.3204	40.3204
7	0.80067	31.9868	23.1564	Material 1	9	32	22.527	30.8899	35.0312	0	35.0312	44.666	44.666
8	0.80067	35.6187	24.8974	Material 1	9	32	23.6199	32.3885	37.4294	0	37.4294	48.3921	48.3921
9	0.80067	38.8688	26.6634	Material 1	9	32	24.4431	33.5173	39.2358	0	39.2358	51.5098	51.5098
10	0.80067	41.7196	28.4571	Material 1	9	32	25.007	34.2906	40.4734	0	40.4734	54.0269	54.0269
11	0.80067	44.151	30.2819	Material 1	9	32	25.3208	34.7209	41.1622	0	41.1622	55.9477	55.9477
12	0.80067	46.1397	32.1413	Material 1	9	32	25.3923	34.819	41.3191	0	41.3191	57.2733	57.2733
13	0.80067	47.6587	34.0395	Material 1	9	32	25.2283	34.594	40.959	0	40.959	58.001	58.001
14	0.80067	48.6764	35.9812	Material 1	9	32	24.8342	34.0537	40.0943	0	40.0943	58.125	58.125
15	0.80067	49.1557	37.9721	Material 1	9	32	24.2149	33.2045	38.7353	0	38.7353	57.6351	57.6351
16	0.80067	49.0524	40.0186	Material 1	9	32	23.3743	32.0518	36.8906	0	36.8906	56.5169	56.5169
17	0.80067	48.3132	42.1285	Material 1	9	32	22.3156	30.6	34.5673	0	34.5673	54.7511	54.7511
18	0.80067	46.8735	44.3114	Material 1	9	32	21.0413	28.8527	31.7709	0	31.7709	52.3124	52.3124
19	0.80067	44.6532	46.579	Material 1	9	32	19.5536	26.8127	28.5063	0	28.5063	49.1685	49.1685
20	0.80067	41.551	48.946	Material 1	9	32	17.8546	24.4829	24.7777	0	24.7777	45.278	45.278
21	0.80067	37.4365	51.4316	Material 1	9	32	15.9464	21.8664	20.5905	0	20.5905	40.5889	40.5889
22	0.80067	32.1357	54.0612	Material 1	9	32	13.8326	18.9678	15.9518	0	15.9518	35.0336	35.0336
23	0.80067	25.4083	56.8704	Material 1	9	32	11.5189	15.7952	10.8747	0	10.8747	28.5247	28.5247
24	0.80067	16.9046	59.9108	Material 1	9	32	9.01644	12.3637	5.38305	0	5.38305	20.944	20.944
25	0.80067	6.07816	63.2638	Material 1	9	32	6.34254	8.69714	-0.484678	0	-0.484678	12.1062	12.1062

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.37466

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30.2919	25.2335	0	0	0
2	31.0926	25.4199	5.58067	0	0
3	31.8933	25.6306	12.0522	0	0
4	32.6939	25.8662	19.0449	0	0
5	33.4946	26.1272	26.2159	0	0
6	34.2953	26.4145	33.2486	0	0
7	35.096	26.7289	39.8527	0	0
8	35.8966	27.0714	45.7637	0	0
9	36.6973	27.443	50.7435	0	0
10	37.498	27.845	54.5816	0	0
11	38.2986	28.279	57.0965	0	0
12	39.0993	28.7465	58.1378	0	0
13	39.9	29.2496	57.5893	0	0
14	40.7006	29.7905	55.3735	0	0
15	41.5013	30.3718	51.4567	0	0
16	42.302	30.9967	45.857	0	0
17	43.1027	31.669	38.6542	0	0
18	43.9033	32.3932	30.0042	0	0
19	44.704	33.1748	20.1582	0	0
20	45.5047	34.0209	9.49041	0	0
21	46.3053	34.9402	-1.46149	0	0
22	47.106	35.9443	-11.9358	0	0
23	47.9067	37.0488	-20.851	0	0
24	48.7073	38.2757	-26.6471	0	0
25	49.508	39.6575	-27.0029	0	0
26	50.3087	41.247	0	0	0

Global Minimum Query (spencer) - Safety Factor: 1.37124

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30.2919	25.2335	0	0	0
2	31.0926	25.4199	6.40087	4.00005	32.0022
3	31.8933	25.6306	13.6732	8.5447	32.0022
4	32.6939	25.8662	21.3756	13.3581	32.0022
5	33.4946	26.1272	29.1219	18.199	32.0023
6	34.2953	26.4145	36.5757	22.857	32.0022
7	35.096	26.7289	43.4464	27.1507	32.0023
8	35.8966	27.0714	49.4865	30.9253	32.0023
9	36.6973	27.443	54.4885	34.0512	32.0023
10	37.498	27.845	58.284	36.4231	32.0023
11	38.2986	28.279	60.7425	37.9594	32.0022
12	39.0993	28.7465	61.771	38.6022	32.0023
13	39.9	29.2496	61.3154	38.3175	32.0023
14	40.7006	29.7905	59.3614	37.0963	32.0022
15	41.5013	30.3718	55.9374	34.9566	32.0022
16	42.302	30.9967	51.1187	31.9453	32.0022
17	43.1027	31.669	45.0325	28.1419	32.0023
18	43.9033	32.3932	37.8665	23.6637	32.0023
19	44.704	33.1748	29.8795	18.6724	32.0022
20	45.5047	34.0209	21.4172	13.3841	32.0022
21	46.3053	34.9402	12.934	8.08273	32.0021
22	47.106	35.9443	5.02629	3.14105	32.0023
23	47.9067	37.0488	-1.5175	-0.948319	32.0022
24	48.7073	38.2757	-5.63622	-3.52221	32.0023
25	49.508	39.6575	-5.85562	-3.65931	32.0022
26	50.3087	41.247	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	95, 20 95, 45 55, 45 30, 25 10, 25 10, 20